

José Henrique Targino Dias Góis

Systems Programmer | Linux | Rust | Cybersecurity

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Profile

Systems programmer and researcher transitioning from computational neuroscience to operating systems and cybersecurity. Experienced with Linux, low-level programming, scientific computing, and data analysis. Currently studying Computer Science and developing Rust-based abstractions for Linux kernel interfaces and userspace environments. Brings a PhD-level research background, peer-reviewed publications, and a strong foundation in quantitative problem solving.

Technical Focus

Systems	Linux, operating systems, Linux kernel interfaces, low-level programming
Programming	Rust, C, C++, Python, Shell, MATLAB
Web	HTML, CSS, JavaScript
Embedded	Arduino, ESP32
Tools and methods	Git, data analysis, scientific computing, experimental automation

Selected Projects

Ample <i>Linux kernel abstraction crate</i>	Rust GitHub profile
Provides Rust abstractions over interfaces exposed by the Linux kernel.	
Userspace <i>Userspace environment</i>	Rust GitHub profile
Bootstraps Rust from assembly and provides a userspace environment for low-level experimentation.	

Experience

Graduate Researcher <i>Instituto do Cérebro, Universidade Federal do Rio Grande do Norte (UFRN)</i>	2012–2020 Natal, Brazil
- Designed, conducted, and analyzed experimental neuroscience studies involving electrophysiology, microscopy, neuronal activity, and behavioral data. - Built reproducible data-analysis workflows using MATLAB and computational methods for extracting and interpreting signals from research datasets. - Contributed to peer-reviewed publications in neuroscience and bibliometrics, working across experimental and computational research.	
Programmer <i>AR Consultoria em Informática</i>	2005–2006 Natal, Brazil
Maintained and developed modules for an enterprise information-management system using Delphi/Pascal and Firebird SQL.	

Education

BSc in Computer Science <i>Universidade Federal do Rio Grande do Norte (UFRN)</i>	In progress; expected 2028 Natal, Brazil
BSc in Information Technology <i>Universidade Federal do Rio Grande do Norte (UFRN)</i>	Completed 2025 Natal, Brazil
PhD in Neuroscience <i>Universidade Federal do Rio Grande do Norte (UFRN)</i>	2016–2020 Computational neuroscience
MSc in Neuroscience <i>Universidade Federal do Rio Grande do Norte (UFRN)</i>	2012–2015 Cellular and molecular neuroscience
BSc in Science and Technology <i>Universidade Federal do Rio Grande do Norte (UFRN)</i>	2009–2012

Selected Publications

- Characterizing Speed Cells in the Rat Hippocampus. *Cell Reports*, 2018.

- **Interconnection and synchronization of neuronal populations in the mouse medial septum/diagonal band of Broca.** *Journal of Neurophysiology*, 2014.
- **Dream characteristics in a Brazilian sample: an online survey focusing on lucid dreaming.** *Frontiers in Human Neuroscience*, 2013.
- **Rising Publication Delays Inflate Journal Impact Factors.** *PLOS ONE*, 2012.

Languages and Academic Recognition

Languages	Portuguese (native), English (professional working proficiency), Spanish (intermediate)
Scholarships	Research funding and scholarships from CNPq, CAPES, CONICET, and UFRN/Propesq